

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

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Queue Application:

- a. Write a program to: a. Implement a queue using an array.

Code:

```
from array import array

queue = array("i")
queue.append(10)
queue.append(20)
queue.append(30)

print("Queue:", queue)

queue.pop(0)

print("After Dequeue:", queue)
print("Himanshu S037")
```

Output:

```
Queue: array('i', [10, 20, 30])
After Dequeue: array('i', [20, 30])
Himanshu S037
```

- b. Simulate a simple queuing system (e.g., customer service queue).

Code:

```
import time
```

```
que = [("Jay.. ", 2), ("Kaushal... ", 3), ("Priyanshi..... ", 5),  
("Amit..... ", 7)]  
  
print("-----Booking ticket-----")  
  
print(que)  
  
while que:  
    job, timer = que.pop(0)  
    print(f"Running:{job}{timer}sec")  
    time.sleep(timer)  
  
print(que)  
  
print("Himanshu S037")
```

Output:

```
Running:Jay.. 2sec  
Running:Kaushal... 3sec  
Running:Priyanshi..... 5sec  
Running:Amit..... 7sec  
[]  
Himanshu S037
```

Curiosity Questions:

1. Where do you see queues in daily life?
2. Why is **FIFO (First In First Out)** fair in real life? Can you think of situations where it's *not* fair?
3. What will happen if you try to **enqueue more elements than the queue size**?
4. What if you **dequeue from an empty queue**?
5. Imagine you're designing a **hospital emergency system**. Should it use a simple queue or something else? (hint: priority queue).
6. If **5 customers arrive at the same time**, but the service desk can serve only 2 at once, how would you design that?